

# Boutir: Enabling merchants to thrive globally with online shop builder powered by App Engine

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## About Boutir

Boutir develops a  
end shop builder,  
100,000+ merchants  
Asia, achieves un  
online service sc  
pandemic, and fo  
driven CRM strate  
Google Cloud.

Boutir is a Hong Kong-based online commerce platform that allows anyone to set up a mobile shop and start selling within minutes, with no technical knowledge required. Its store builder handles ordering to checkout, delivery, and end-client notifications automatically. Since launching in 2014 with a social enterprise vision of merchant empowerment, Boutir has spread across Asia to markets including Malaysia and Indonesia serving more than 100,000 merchants. It has won multiple awards including the Hong Kong Retail Innovation Award for Best Retail

## Google Cloud results

- Enables mission to empower merchants by supporting growth from under 10 HK merchants to 100,000+ across Asia
- Promotes creative experimentation by enabling simultaneous A/B testing with zero downtime
- Enhances customer loyalty by seamlessly absorbing order surge to 1,000s per second without issues

10x

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Innovation Product and the HSBC Youth Business Award Merit Prize.

Industries: Technology

Location: Hong Kong

Cloud Storage

Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

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Cloud Storage (<https://cloud.google.com/storage/>)

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Hong Kong startup innovator Eric Ng built a successful career helping other digital entrepreneurs succeed, building their apps from his platform

RedSo (<https://www.redso.com.hk/about>). At the same

time, Ng yearned to launch his own application carrying a personal stamp, while advancing his mission of empowering fellow entrepreneurs.

That led to the creation of Boutir

(<https://www.boutir.com/HK/en/>), an end-to-end online shop builder that allows any merchant in the markets it operates in to set up a mobile shop within minutes. On Boutir, anybody can download the app, register for free, and obtain a URL to a functional shopfront. All the merchant needs to do next is upload a product and an online shop is born.

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*—Eric Ng, co-founder and  
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Boutir links to major payment providers and delivery services, and offers its own inventory management solution. That means everything from ordering to checkout, delivery, and end-client notifications is handled automatically. Since 2014, Boutir has spread from Hong Kong across Asia to markets including Malaysia and Indonesia serving more than 100,000 merchants.

As a seasoned developer, Ng felt that only a trusted cloud provider could deliver the seamless development experience, real-time data analytics, and unlimited scalability needed for success. He chose Google Cloud (<https://cloud.google.com/>), and soon saw merchants from across Asia flocking to Boutir to make their ecommerce dreams take flight.

"Our mission is to empower anybody to launch a digital store with just a mobile phone," says Ng. "For such a simple system to work, there's much

complex development needed on an agile cloud infrastructure. Google Cloud enables both Boutir and its clients to grow at a dizzying pace with zero downtime."

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Even with thousands placing orders per second, none of Boutir's merchants experienced crashes, thanks to App Engine's scalability. They were able to meet an important social need, while also winning

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## **Absorbing massive pandemic demand spikes with App Engine resilience**

Ng's experience with Google Cloud goes back to the Google platform's inception in 2008, when he began building projects on App Engine (/appengine) during a time the app development solution was still in beta phase. Even back then, he grasped the freedom that allowed a developer to focus 100% on app building, leaving infrastructure, scaling, and connectivity entirely to Google Cloud.

Ng's RedSo business grew and evolved in step with App Engine, and he came to know firsthand how its ease-of-use combined with unlimited scalability to provide an optimal architecture for startup developers like himself. When the time came to launch Boutir, App Engine was the natural choice for development and deployment, since the mission was to scale to a global level. The decision was rewarded with rapid success.

However, despite knowing the capabilities of App Engine, Ng was surprised by the product's resilience

when the COVID-19 crisis triggered a scramble for life-protecting face masks among Hong Kong's 7.5 million population. Stores saw lines of up to three hours to buy the items, while even major online merchants saw systems crash due to the high demand.

"Hong Kongers were eager to protect their loved ones with life-saving masks in the pandemic," says Ng. "Even with thousands placing orders per second, none of Boutir's merchants experienced crashes thanks to App Engine's scalability. They were able to meet an important social need, while also winning customer loyalty through reliability."

"With App Engine, we can deploy several new versions in hot deployment with no need to take down the systems. That's so empowering because it means we can experiment in new markets

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## **Testing global markets with hot deployments enabled by App Engine**

For Ng, one of the key benefits of App Engine is the ability to trigger simultaneous hot deployments of new app features. Hot deployments are new app feature launches that don't require stopping the server instance and restarting. This means that a platform can offer a continuous stream of innovation with zero service interruptions.

Alongside continuous app improvement, hot deployments enabled by App Engine play a key role in Boutir's global expansion plans. They allow the platform to launch different app versions in any new market and gauge feedback on each one, for real-time, multi-version A/B testing, which randomly compares two variants in the same market to find out which one is more popular.

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The continuously improving ability of App Engine to support multiple hot deployments also enables Boutir to accelerate app development cycles. Compared to platforms hosted on bare metal, Ng estimates that hot deployment on App Engine provides a 10-fold speed advantage in deploying new application features. But more importantly, when there are specific incidents to address on the back end, hot deployment can save the business by reducing business downtime to zero. This makes the development environment more engineer friendly, which improves the quality and level of creativity as a whole.

The ability to roll back to a deployment at any time in the event of a glitch, to the precise point of a coding error, brings a priceless intangible asset to the team. If anything goes wrong, Ng explains, App Engine allows Boutir to immediately switch back to the old version while pinpointing and correcting the coding problem.

"Knowing that with App Engine we have the freedom to make mistakes, and can just roll back and fix the problem, gives us incredible confidence and peace of mind to be as creative as we want adding new features to our platform," says Ng.



## **A vibrant future enabling the world's merchants with BigQuery and Google AI solutions**

Boutir deploys BigQuery (/bigquery), Google Cloud's data warehousing solution, for crunching the reams of data that pour into its systems to boost performance. The solution has proved crucial for Boutir to obtain data insights on an intuitive dashboard that drive overall platform improvements which translate into improved client sales.

The next step is to deploy BigQuery to provide Boutir clients with their own business intelligence, enabling them to offer targeted product recommendations through granular sales trend analysis. Ng says customer relations management (CRM), enabled by BigQuery, promises to become a game changer in enabling even the smallest mom-and-pop establishments to maximize their client base, by giving insight into customer preferences.

Meanwhile, Boutir foresees adopting Vision AI (/vision) to analyze product photos which are currently stored and dynamically resized on Cloud

Storage (/storage). This will enable automated categorization that eliminates the need for merchants to manually enter merchandise information for their website. The result will be better efficiencies, happier customers, and ultimately cost savings passed on to consumers.

## **Ambitious push into exciting Southeast Asian market powered by Kubernetes**

These AI-driven Google Cloud solutions will facilitate Boutir's immediate strategy to drive a push into the Southeast Asian market, which it sees as exciting due to a young Internet-savvy population that prefers mobile interactions.

Due to the popularity of livestreaming in ecommerce markets such as Vietnam and Thailand, Boutir is considering ways to further lower network latency and improve scalability. That is driving a move to adopt containerized solutions on Google Kubernetes Engine (/kubernetes-engine), which provide more powerful, granular, and cost-effective scalability.

"We have an open-ended vision of our journey empowering global merchants, starting with our major push into Southeast Asia," says Ng. "We're inspired by the constantly evolving solutions of Google Cloud that bring us open-ended possibilities to create our future."

